

**RAMCO INSTITUTE OF TECHNOLOGY**  
**Department of Electronics and Communication Engineering**  
**Academic Year: 2018 - 2019 (Odd Semester)**

**Innovative Practices Description**

**UNIT I SAMPLING & QUANTIZATION**

**Degree, Semester & Branch:** V Semester B.E. ECE B

**Course Code & Title:** EC6501 Digital Communication

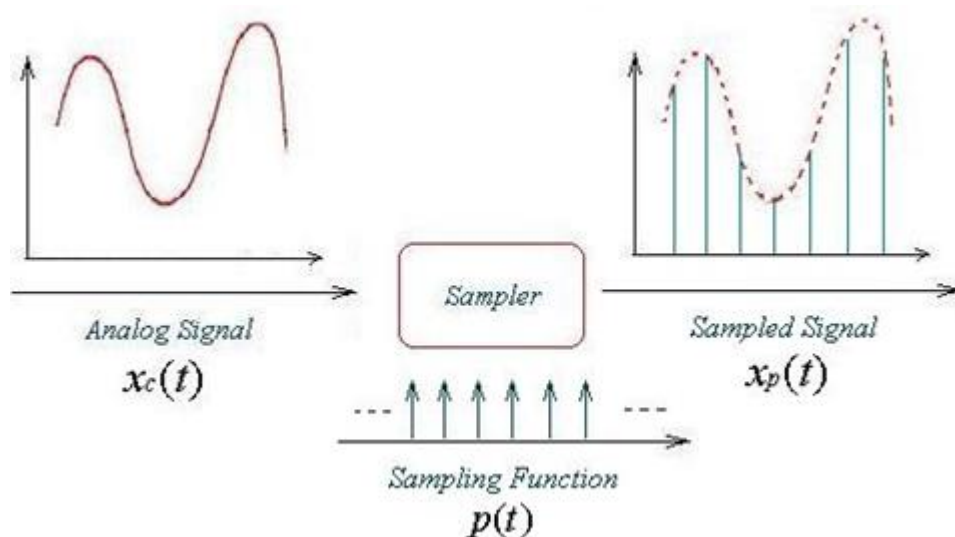
**Name of the Faculty member:** Mrs.G.Gnana Priya

**Name of the Topic:** Sampling

**Name of the Innovative Practice:** Simulation (Linked with Lab Experiments)

**Description:**

Generally signals are analog in nature eg., speech, weather signals. To process the analog signal by digital means, it is essential to convert them to discrete time signal, and then convert them to a sequence of numbers. The process of converting an analog to digital signal is 'Analog to Digital Conversion'. The ADC involves three steps which are sampling, quantization and coding. Matlab simulation was shown to the students for better understanding about the sampling process.



| Course Outcome | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|----------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| CO1            | 2   | 2   | 3   |     | 2   |     |     | 1   |     | 2    |      | 2    |

*CO1: The students will be able to design applications using the various types of sampling and quantization*

**References:**

- S.Haykin, "Digital Communications", John Wiley, 2005
- J.G Proakis, "Digital Communication", 4th Edition, Tata Mc Graw Hill Company, 2001.